

A Minimal Three-Dimensional Counterexample for Eventual Cone Invariance

A candidate full answer to the question after Example 3.1 of arXiv:2303.07809

Autonomous counterexample candidate for human review

11 August 2026

Status. Candidate full counterexample and minimal-dimension resolution, likely valid. An explicit nilpotent 3×3 matrix is individually but not uniformly eventually nonnegative on the 2×2 positive-semidefinite cone. A bonus restriction argument also improves a secondary dimension bound from 16 to 6.

1 Source question

Glück and Hölz [1] construct a four-dimensional ordered space and a matrix semigroup that is individually but not uniformly eventually nonnegative. Immediately after the construction they ask whether a counterexample exists in dimension three.

$$e^{t_n A} x(n) = \begin{pmatrix} 1 \\ 2\sqrt{2n-1} \\ 0 \end{pmatrix}.$$

Using this explicit formula, one can now check by a brief computation that the vector $z := e^{t_n A} x(n)$ satisfies $2z_1 z_2 < z_2^2 + z_3^2 + z_4^2$, so $e^{t_n A} x(n) \notin \mathbb{R}_+^4$. Hence, we found a sequence of vectors $x(n)$ in $\mathbb{R}_+^4$ and a sequence of times $t_n \rightarrow \infty$ such that $e^{t_n A} x(n) \notin \mathbb{R}_+^4$ for each n . So $(e^{tA})_{t \geq 0}$ is not uniformly eventually nonnegative.

We do not know whether there also exists a counterexample in dimension 3.

Remark 3.2. In [19, Proposition 1] it was claimed that individual and uniform eventual nonnegativity are equivalent.² Example 3.1 shows that this is not correct.³

At first glance this appears to cause complications for some of the arguments in [19] that rely on spectral results for eventually nonnegative matrix semigroups from [12], as the latter were established in [12] only for the case of uniform eventual nonnegativity. This can be resolved, though, as we show in Section 4 that the same spectral results remain true for the individually (and in fact, even the weakly) eventually nonnegative case.

Figure 1: The dimension-three question after Example 3.1, source PDF page 7.

Recall that individual eventual nonnegativity means that for each x in the cone there is a time $t_0(x)$ such that $e^{tA}x$ stays in the cone for every $t \geq t_0(x)$. Uniform eventual nonnegativity requires one common t_0 for all cone vectors.

2 The three-dimensional counterexample

Let

$$\mathcal{K} = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 \geq 0, x_3 \geq 0, x_2^2 \leq x_1 x_3\}. \quad (1)$$

Under the identification

$$(x_1, x_2, x_3) \longleftrightarrow \begin{pmatrix} x_1 & x_2 \\ x_2 & x_3 \end{pmatrix},$$

this is the positive-semidefinite cone. It is therefore a closed, pointed, convex cone with nonempty interior.

Set

$$A = \begin{pmatrix} 0 & 4 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}. \quad (2)$$

Since $A^3 = 0$,

$$e^{tA} = I + tA + \frac{t^2}{2}A^2 = \begin{pmatrix} 1 & 4t & 2t^2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}. \quad (3)$$

Theorem 1. *The semigroup $(e^{tA})_{t \geq 0}$ is individually eventually nonnegative with respect to $\mathcal{K}$, but is not uniformly eventually nonnegative. Indeed, $e^{tA}\mathcal{K} \not\subseteq \mathcal{K}$ for every $t > 0$.*

Proof. Fix $x = (x_1, x_2, x_3) \in \mathcal{K}$ and write $y = e^{tA}x$. Formula (3) gives

$$y_1 = x_1 + 4tx_2 + 2t^2x_3, \quad y_2 = x_2 + tx_3, \quad y_3 = x_3.$$

A direct expansion yields the determinant identity

$$y_1 y_3 - y_2^2 = (x_1 x_3 - x_2^2) + 2tx_2 x_3 + t^2 x_3^2. \quad (4)$$

If $x_3 = 0$, the cone inequality forces $x_2 = 0$, so $e^{tA}x = x$ for every $t \geq 0$. Suppose $x_3 > 0$. Since the first term on the right of (4) is nonnegative, the whole expression is nonnegative whenever

$$t \geq t_0(x) := \max \left\{ 0, -\frac{2x_2}{x_3} \right\}.$$

For these times $y_3 = x_3 > 0$, and the determinant inequality also implies $y_1 \geq y_2^2/y_3 \geq 0$. Thus $e^{tA}x \in \mathcal{K}$ for every $t \geq t_0(x)$. This proves individual eventual nonnegativity.

Now fix an arbitrary $t > 0$ and take

$$x(t) = (t^2, -t, 1). \quad (5)$$

It lies on the boundary of $\mathcal{K}$, because $(-t)^2 = t^2 \cdot 1$. But

$$e^{tA}x(t) = (-t^2, 0, 1) \notin \mathcal{K}.$$

Hence e^{tA} is not cone-preserving at any positive time, which rules out uniform eventual nonnegativity. $\square$

Proposition 1. *Dimension three is minimal for an individual-versus-uniform separation.*

Proof. A closed pointed cone with nonempty interior in a two-dimensional real vector space is generated by its two boundary rays, so it is polyhedral. If u, v generate the cone and a semigroup is individually eventually nonnegative, choose eventual times t_u, t_v for those two vectors. For $t \geq \max\{t_u, t_v\}$ and every $\alpha, \beta \geq 0$,

$$e^{tA}(\alpha u + \beta v) = \alpha e^{tA}u + \beta e^{tA}v$$

is again in the cone. Thus the eventual time is uniform. The same is immediate in dimension one. Theorem 1 therefore occurs in the smallest possible dimension. $\square$

3 Bonus: weak versus individual in dimension six

The source derives a weakly but not individually eventually nonnegative semigroup on the 16-dimensional operator space over its four-dimensional example. Theorem 1 first reduces that construction to dimension 9. It can be restricted further.

Let $X = \mathbb{R}^3$ with cone $\mathcal{K}$, and let $\mathcal{C} = \mathcal{L}(X)_+$ denote the cone of $\mathcal{K}$ -preserving operators. Choose

$$u = (1, 0, 1) \in \text{int } \mathcal{K}, \quad \varphi(x) = x_1 + x_3 \in \text{int } \mathcal{K}', \quad U = u \otimes \varphi.$$

Thus $U(\mathcal{K} \setminus \{0\}) \subset \text{int } \mathcal{K}$, so U is an interior point of $\mathcal{C}$. Define

$$Y = \text{span}\{I, A, A^2, U, AU, A^2U\}, \quad Y_+ = Y \cap \mathcal{C}. \quad (6)$$

The six displayed matrices are linearly independent: in matrix-unit notation,

$$\begin{aligned} A &= 4E_{12} + E_{23}, & A^2 &= 4E_{13}, & U &= E_{11} + E_{13} + E_{31} + E_{33}, \\ AU &= E_{21} + E_{23}, & A^2U &= 4E_{11} + 4E_{13}, & I &= E_{11} + E_{22} + E_{33}. \end{aligned}$$

Their unique matrix-unit entries successively force all coefficients in a linear relation to vanish. Hence $\dim Y = 6$. Moreover Y_+ is a closed cone with nonempty relative interior because it contains the ambient interior point U .

Let $B : Y \rightarrow Y$ be left multiplication by A . The space is B -invariant, since the two triples in (6) are nilpotent A -chains, and

$$e^{tB}M = e^{tA}M. \quad (7)$$

Proposition 2. *The six-dimensional ordered-space semigroup $(e^{tB})_{t \geq 0}$ is weakly but not individually eventually nonnegative.*

Proof. Because Y contains an interior point of $\mathcal{C}$, the finite-dimensional positive-extension theorem extends every positive functional on (Y, Y_+) to a positive functional on $(\mathcal{L}(X), \mathcal{C})$. The latter dual cone is the conical hull of the evaluation functionals

$$M \mapsto \langle x', Mx \rangle, \quad x \in \mathcal{K}, \quad x' \in \mathcal{K}';$$

this is Proposition 2.3 of [1].

Fix $M \in Y_+$. For each such evaluation, $Mx \in \mathcal{K}$, so Theorem 1 gives

$$\langle x', e^{tB}Mx \rangle = \langle x', e^{tA}Mx \rangle \geq 0$$

for all sufficiently large t . A positive functional is a finite conical combination of evaluations, so one common maximum of their individual times proves weak eventual nonnegativity on Y .

On the other hand $I \in Y_+$, while $e^{tB}I = e^{tA}I \notin \mathcal{C}$ for every $t > 0$ by (5). Thus the positive vector I never has an eventually nonnegative orbit in Y , and the semigroup is not individually eventually nonnegative. $\square$

No claim is made that six is the smallest dimension for this second separation.

4 Verification and novelty status

The main construction uses only a nilpotent exponential and one quadratic cone inequality. A bounded search of the run indexes, exact phrases from the source, the journal citation page, and later web/arXiv records found no prior dimension-three construction. Priority is not asserted; both results should receive specialist review before dissemination.

References

- [1] J. Glück and J. Hölz, *Eventual cone invariance revisited*, Linear Algebra Appl. 675 (2023), 274–293; arXiv:2303.07809.